

A certified $4.1357418449\times$ resource reduction

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For the unchanged periodic 12×12 spin- $\frac{1}{2}$ isotropic Heisenberg benchmark at $T = 1$ and global operator-norm tolerance 10^{-6} , a new post-freeze certificate accepts 95 fourth-order Suzuki steps and rejects 94. The published rigorous control uses 393 steps. Under the same compilation model, merged group exponentials decrease from 11,791 to 2,851, giving the exact ratio $11791/2851 = 4.135741844966678\dots$. The improvement integrates an exact, SHA-256-bound degree-five Pauli sidecar into the finite-step error ledger. Fast verification, deep coefficient regeneration, adjacent-step minimality, and corruption-rejection tests all pass. No fivefold result is claimed.

The Hamiltonian, $L = 12$, $N = 144$, periodic boundary conditions, time, error metric, four-matching decomposition, 31-stage five-copy fourth-order Suzuki formula, and compiled-cost rules are identical on both sides. Adjacent stage merging gives $G(r) = 30r + 1$ group exponentials. Each matching contains 72 bonds and the common upper-bound model charges three CNOTs per bond propagator.

Table 1: Integer resource accounting under one unchanged cost model.
$$\frac{G(393)}{G(95)} = \frac{11791}{2851} = 4.13574184496667835847\dots$$

The frozen schema-v3 result used a verified anticommuting D4 partition but a generic D5 majorant. The new certificate keeps the D4 proof unchanged and replaces only that D5 majorant with an exact grouped sidecar. It contains 605,832 canonical Pauli coefficients partitioned into 123,106 same-support, pairwise-anticommuting groups of size at most ten. Its exact site-norm upper bound is

For each group $K_g = \sum_j c_j P_j$, pairwise anticommutation permits the outward rational bound

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The grouping search is untrusted. The verifier independently checks payload canonicity and SHA-256, exact coverage and absence of duplicates, support equality, every symplectic anticommutation relation, every outward square-root bound, and all metadata copied into the main certificate.

3 Finite-step ledger and adjacent boundary

Right-generator contribution at $r = 95$	Global upper bound
D4	$5.7218800760132647 \times 10^{-7}$
D5, exact grouped	$3.4853484615103630 \times 10^{-8}$
D6	$1.7893140877453180 \times 10^{-7}$
D7	$2.1921361303553973 \times 10^{-8}$
D8 and above, geometric tail	$1.8255487798872918 \times 10^{-7}$
Total	$9.9044914028324506 \times 10^{-7}$

Table 2: Outward-readable decimals projected from exact rational values.

All acceptance decisions use exact rational arithmetic, not the displayed decimals. At the adjacent step $r = 94$, the same function evaluates to $1.0468061165603706 \times 10^{-6}$, above the tolerance. Hence 95 is the smallest integer accepted by this certificate. Fast verification now regenerates each displayed contribution and the adjacent-step value. In particular, setting D6, D7, or the tail to zero and synchronously recomputing the submitted total is rejected.

4 Verification and reproducibility

Fast verification is suitable for a live demonstration:

```
PYTHONPATH=src python3 scripts/verify.py \
  certificates/issue128-d5-integrated-certificate.json
```

It returns `valid=true`, `candidate_steps=95`, `candidate_group_exponentials=2851`, `d5_term_count=605832`, and the exact ratio `11791/2851`. It also returns `finite_step_bound_recomputed=true`. Deep verification is invoked with `-deep`. It scans all 31 published-theorem centers, selects center 20, regenerates all D4 and D5 coefficients from the Suzuki formula, and requires exact interval equality with the two sidecars. The current-source deep run took 408.88 seconds, used at most 1,538,768,896 bytes of resident memory, and returned `deep_proof_regenerated=true` and `baseline_centers_scanned=31`.

A second verifier uses only the Python standard library:

```
python3 scripts/reference_verify.py \
  certificates/issue128-d5-integrated-certificate.json
```

It independently checks sidecar coverage, symplectic anticommutation, square-root domination, and equality of three D4/D5 metadata values: term count, group count, and maximum group size. It also checks every finite-step contribution, the Suzuki tail, the 95/94 comparison, and resource arithmetic in 3.63 seconds. It does not regenerate the coefficient maps; that separate obligation belongs to deep mode. The normal suite reports 146 passed tests and 12 explicitly deselected slow tests; ten focused certificate mutations and eight independent-checker tests pass.

On a nondegenerate open 2×3 diagnostic with seven unique bonds, dense evaluation at 95 steps gives operator-norm error $1.8313133889571672 \times 10^{-10}$, below the outward scaled periodic comparison $4.129166666666667 \times 10^{-8}$. This checks normalization and stage ordering; it is not a proof that the periodic certificate applies to open boundaries.

The original ten-file delivery manifest remains frozen at 97 steps and 11791/2911. It has not been silently rewritten. The new artifact `issue128-d5-integrated-certificate.json` is separately reproducible from the old certificate and the existing D5 sidecar using `scripts/build_d5_integrated_certificate.py`.

5 Fivefold boundary

Fivefold resource arithmetic requires $r \leq 78$. However, the current D4 contribution alone is approximately 1.259084×10^{-6} at $r = 78$, already larger than the entire tolerance. Nonnegative D5-tail contributions can only increase the ledger. Exact-D8 HPC work therefore remains diagnostic: it may guide a future proof, but it cannot establish fivefold without a new D4 tightening. This report makes no fivefold claim.

6 Claim boundary and release gates

This certificate is scoped to the stated Hamiltonian, normalization, periodic boundary, formula, tolerance, and cost model. It does not prove that 94 physical steps fail, that the grouping heuristic is optimal, or that the method is state-of-the-art over unrestricted quantum simulation algorithms.

A release is accepted only when:

- the 95-step primary and standard-library reference verifiers pass;
- deep mode reports 31 scanned centers and coefficient regeneration;
- all ordinary and adversarial mutation tests pass;
- the D5-integrated SHA-256 manifest is clean; and
- this PDF is rebuilt from the committed $\text{\LaTeX}$ source and visually inspected page by page.